

Manny's Camera Subsystem Power Budget

Team Number:	303
Project Name:	Sable Camera Subsystem
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Version:	1

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators, resistors, capacitors, or passive elements							
All Major Components	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	ESP32-S3	ESP32-S3-WROOM-1-N4	+3 - 3.6V	1	500	500	mA
	Yellow LED	LTST-C190TBKT	+3.3V	1	20	20	mA
	Green LED	LTST-C190TBKT	+3.3V	2	20	40	mA
	Red LED	LTST-C190TBKT	+3.3V	2	20	40	mA
	OV2640 Camera Sensor AVDD	OV2640	+2.8-3.3V	1	100	100	mA
	OV2640 Camera Sensor DOVDD	OV2640	+1.7-2.8V	1	100	100	mA
	OV2640 Camera Sensor DVDD	OV2640	+1.2-1.5V	1	30	30	mA
	3.3 Voltage Regulator	LM2575D2T-3.3R4G	+3.3V	1	350	350	mA
	2.8 Voltage Regulator	TPS7A0328PDBVR	+1.5-6V	1	200	200	mA
	1.5 Voltage Regulator	TLV70015DDCR	+1.5-5.5V	1	200	200	mA
B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in the design.							
+3.3V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	ESP32-S3	ESP32-S3-WROOM-1-N4	+3 - 3.6V	1	500	500	mA
	Yellow LED	LTST-C190TBKT	+3.3V	1	20	20	mA
	Green LED	LTST-C190TBKT	+3.3V	2	20	40	mA
	Red LED	LTST-C190TBKT	+3.3V	2	20	40	mA
	OV2640 Camera Sensor AVDD	OV2640	+2.8-3.3V	1	100	100	mA
	Subtotal					700	mA
	Safety Margin					25%	
	Total Current Required on +3.3V Rail					875	mA
c2. Regulator or Source Choice	+3.3V Switching Regulator	LM2575D2T-3.3R4G	+3.3V	1	1000	1000	mA
	Total Remaining Current Available on +5V Rail					125	mA
+2.8 Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	OV2640 Camera Sensor DOVDD	OV2640	+1.7-2.8V	1	100	100	mA
	Subtotal					100	mA
	Safety Margin					25%	
	Total Current Required on +2.8V Rail					125	mA
c3. Regulator or Source Choice	+2.8V LDO Regulator	TPS7A0328PDBVR	+1.5-6V	1	200	200	mA
	Total Remaining Current Available on -5V Rail					75	mA
+1.5V Power Rail	Component Name	Part Number	SupplyVoltageRange	#	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
	OV2640 Camera Sensor DVDD	OV2640	+1.2-1.5V	1	30	30	mA
	Subtotal					30	mA
	Safety Margin					25%	
	Total Current Required on +1.5V Rail					37.5	mA
c4. Regulator or Source Choice	+1.5V LDO Regulator	TLV70015DDCR	+1.5-5.5V	1	200	200	mA
	Total Remaining Current Available on 3.3V Rail					162.5	mA
C. For each power rail above, select a specific voltage regulator using the same process as for major component selection. Confirm that the Total Remaining Current Available on each rail above is not negative.							
D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of the regulators for all of the power rails simultaneously. If you need multiple power							
External Power Source 1	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	0930	110VAC	+9V	3000	3000	mA
Power Rails Connected to External Power Source 1	+3.3V Switching Regulator	LM2575D2T-3.3R4G	+3.3V	1	1000	1000	mA
						0	mA
						0	mA
	Total Remaining Current Available on External Power Source 1					2000	mA
External Power Source 2	Component Name	Part Number	SupplyVoltageRange	Output	AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
Power Source 2 Selection	Shared 12V DC 8 pin connector	(full part number)	+12V	-9V	1000	1000	mA
Power Rails Connected to External Power Source 2	+3.3V Switching Regulator	LM2575D2T-3.3R4G	+3.3V	1	1000	1000	mA
	Total Remaining Current Available on External Power Source 2					0	mA

Notes

External Supply Voltage should be determined by the dropout voltage for highest-voltage regulator (e.g., +14V for a +12V regulator).
If you have multiple units in your design (e.g., a base unit and remote unit) then you need a separate power budget for each unit